

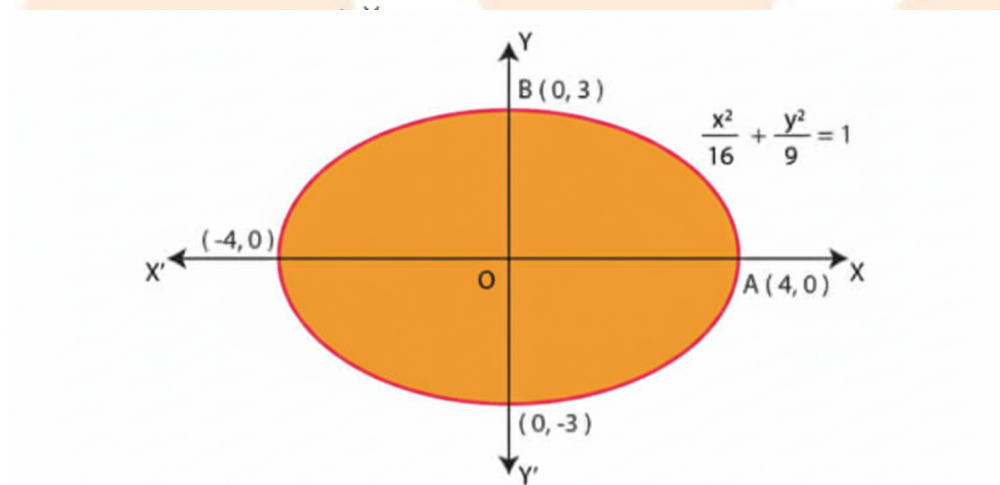
NCERT Solutions for Class 12 Maths

Chapter 8 – Application of integral

Exercise 8.1

1. Find the area of the region bounded by the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

Ans:



Area Bounded by Ellipse

Area of ellipse $= 4 \times$

Area of OAB

Area of OAB $= \int_0^4 y dx$

Substitute $y = 3\sqrt{1 - \frac{x^2}{16}}$

$$= \int_0^4 3\sqrt{1 - \frac{x^2}{16}} dx$$

simplifying,

$$= \frac{3}{4} \int_0^4 \sqrt{16 - x^2} dx$$

$$= \frac{3}{4} \left[\frac{x}{2} \sqrt{16 - x^2} + \frac{16}{2} \sin^{-1} \frac{x}{4} \right]_0^4$$

Substituting the limits,

$$= \frac{3}{4} \left[2\sqrt{16 - 16} + 8\sin^{-1}(1) - 0 - 8\sin^{-1}(0) \right]$$

$$= \frac{3}{4} \left[\frac{8\pi}{2} \right]$$

$$= \frac{3}{4} [4\pi]$$

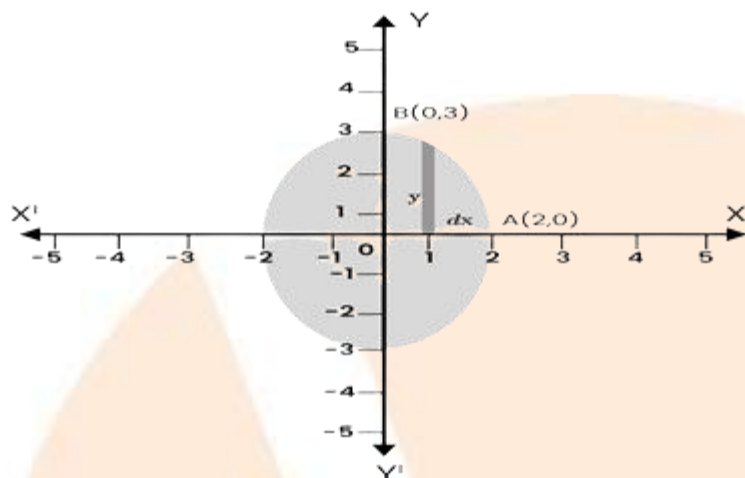
$$= 3\pi$$

As a result, the ellipse's area is

$$= 4 \times 3\pi = 12\pi.$$

2. Find the area of the region bounded by the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

Ans:



Area Bounded by the Ellipse

Area of ellipse $= 4 \times$

Area of OAB

$$\text{Area of OAB} = \int_0^2 y dx$$

$$\text{Substitute } y = 3\sqrt{1 - \frac{x^2}{4}}$$

$$= \int_0^2 3\sqrt{1 - \frac{x^2}{4}} dx$$

simplifying,

$$= \frac{3}{2} \int_0^2 \sqrt{4 - x^2} dx$$

$$= \frac{3}{2} \left[\frac{x}{2} \sqrt{4 - x^2} + \frac{4}{2} \sin^{-1} \frac{x}{2} \right]_0^2$$

Substituting the limits,

$$= \frac{3}{2} \left[\frac{2\pi}{2} \right]$$

$$= \frac{3\pi}{2}$$

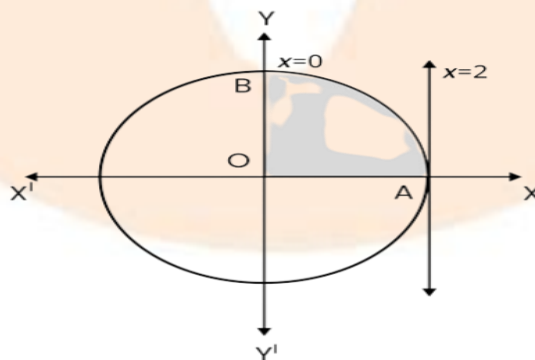
As a result, the ellipse's area is

$$4 \times \frac{3\pi}{2} = 6\pi.$$

3. Area lying in the first quadrant and bounded by the circle $x^2 + y^2 = 4$ and the lines $x=0$

- A. π
- B. $\frac{\pi}{2}$
- C. $\frac{\pi}{3}$
- D. $\frac{\pi}{4}$

Ans:



Area Lying in the First Quadrant and Bounded by the Circle

$$\text{Area OAB} = \int_0^2 y \, dx \text{ substitute } y = \sqrt{4-x^2}$$

$$= \int_0^2 \sqrt{4-x^2} \, dx$$

$$= \left[\frac{x}{2} \sqrt{4-x^2} + \frac{4}{2} \sin^{-1} \frac{x}{2} \right]_0^2$$

Substituting the limits,

$$= 2 \left(\frac{\pi}{2} \right)$$

$$= \pi$$

As a result, the required area is π option A.

4. Area of the region bounded by the curve $y^2 = 4x$, y-axis and the line $y=3$ is

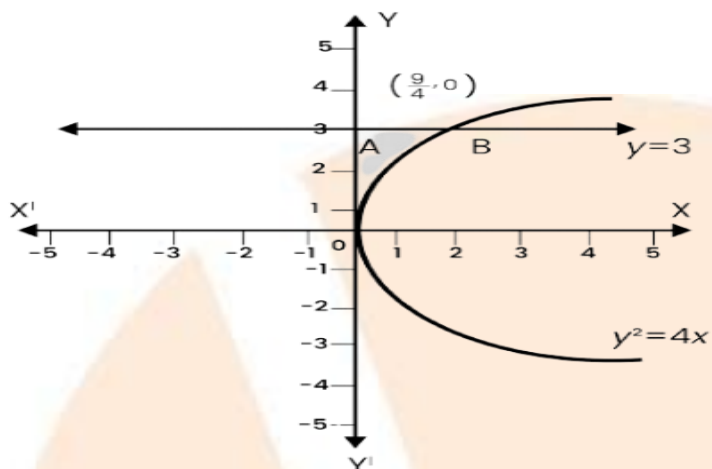
A. 2

B. $\frac{9}{4}$

C. $\frac{9}{3}$

D. $\frac{9}{2}$

Ans:



$$\text{Area OAB} = \int_0^3 x dy$$

$$\text{substitute } x = \frac{y^2}{4}$$

$$= \int_0^3 \frac{y^2}{4} dy$$

$$= \frac{1}{4} \left[\frac{y^3}{3} \right]_0^3$$

Substituting the limits,

$$= \frac{1}{12} (27)$$

$$= \frac{9}{4} \text{ units}$$

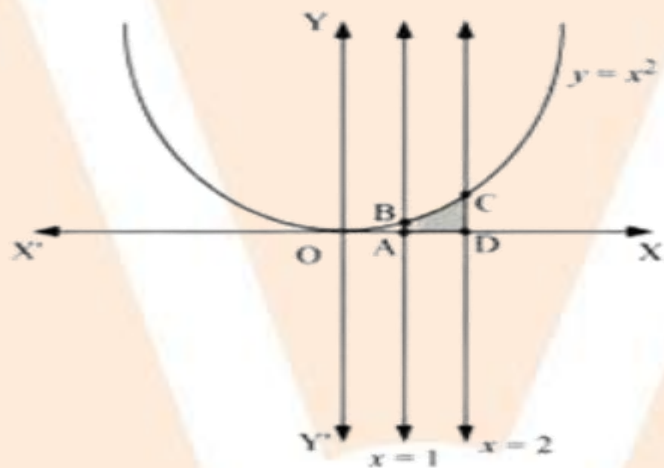
As a result, the correct response is $\frac{9}{4}$ units option B.

Miscellaneous Exercise

1. Find the area under the given curves and given lines:

(i) $y=x^2, x=1, x=2$ and x -axis

Ans:



$$\text{Area ADCBA} = \int_1^2 y dx$$

substitute $y = x^2$

$$= \int_1^2 x^2 dx$$

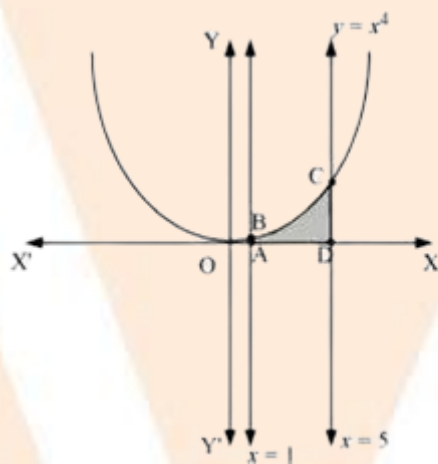
Substituting the limits,

$$= \frac{8}{3} - 0 \frac{1}{3}$$

$$= \frac{7}{3} \text{ units}$$

(ii) $y=x^4, x=1, x=5$ and x -axis

Ans:



$$\text{Area of ADCBA} = \int_1^5 x^4 dx$$

Integrating using the power rule,

$$= \left[\frac{x^5}{5} \right]_1^5$$

Substituting the limits,

$$= \frac{(5)^5}{5} - 0 \frac{1}{5}$$

Simplifying,

$$= (5)^4 - 0 \frac{1}{5}$$

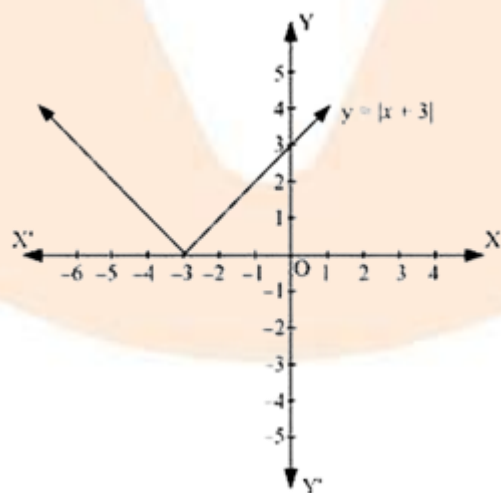
$$= 625 - \frac{1}{5}$$

$$= 624.8 \text{ units}$$

2. Sketch the graph of $y = |x+3|$ and evaluate $\int_{-6}^0 |x+3| dx$.

Ans:

x	-6	-5	-4	-3	-2	-1	0
y	3	2	1	0	1	2	3



$$(x+3) \leq 0 \text{ for}$$

$$-6 \leq x \leq -3$$

$$(x+3) \geq 0$$

$$\text{for } -3 \leq x \leq 0$$

Therefore,

$$\int_{-6}^0 |(x+3)| dx = -\int_{-6}^{-3} (x+3) dx + \int_{-3}^0 (x+3) dx$$

Integrating using the power rule

$$= -\left[\frac{x^2}{2} + 3x\right]_{-6}^{-3} + \left[\frac{x^2}{2} + 3x\right]_{-3}^0$$

Substituting the limits,

$$= -\left[\left(\frac{(-3)^2}{2} + 3(-3)\right) - \left(\frac{(-6)^2}{2} + 3(-6)\right)\right] + \left[0 - \left(\frac{(-3)^2}{2} + 3(-3)\right)\right]$$

Simplifying,

$$= -\left[0\frac{9}{2}\right] - \left[0\frac{9}{2}\right]$$

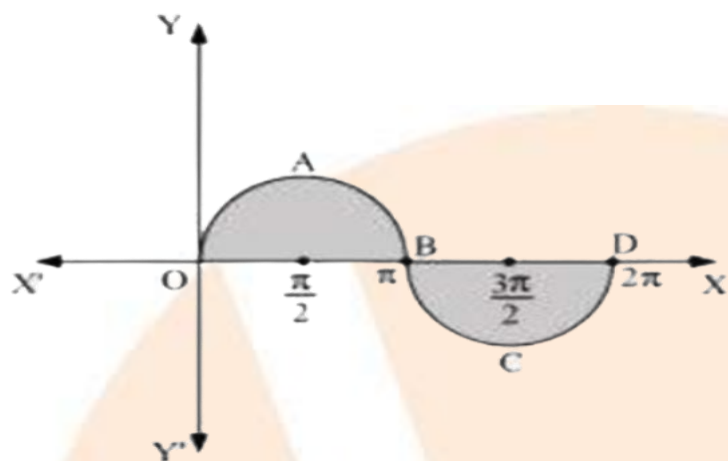
$$= 9$$

3. Find the area bounded by the curve $y = \sin x$ between $x=0$ and $x=2\pi$.

Ans:

Therefore,

$$\text{area} = \text{Area OABO} + \text{Area BCDB}$$



Area Bounded by curve $y = \sin x$

$$= \int_0^{\pi} \sin x dx + \left| \int_{\pi}^{2\pi} \sin x dx \right|$$

$$= [-\cos x]_0^{\pi} + \left| [-\cos x]_{\pi}^{2\pi} \right|$$

Substituting the limits,

$$= [-\cos \pi + \cos 0] + |-\cos 2\pi + \cos \pi|$$

Simplifying,

$$= 1 + 1 + |(-1 - 1)|$$

$$= 2 + |-2|$$

$$= 2 + 2$$

$$= 4 \text{ units}$$

4. Area bounded by the curve $y = x^3$, the x-axis and the ordinates $x = -2$ and $x = 1$ is

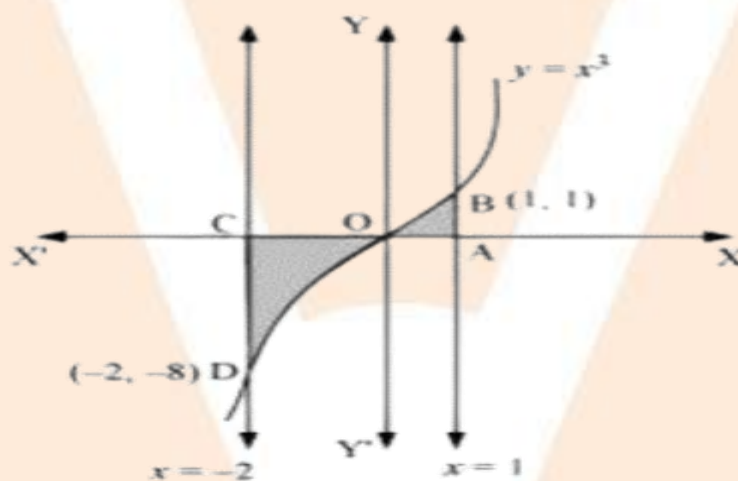
A. -9

B. $0 \frac{15}{4}$

C. $\frac{15}{4}$

D. $\frac{17}{4}$

Ans:



As shown in the diagram, the required area is:

$$\text{Required area} = \int_{-2}^1 y dx$$

$$= \int_{-2}^1 x^2 dx$$

Integrating using the power rule

$$= \left[\frac{x^3}{3} \right]_{-2}^1$$

Substituting the limits,

$$= \left[\frac{1}{3} - \frac{(-2)^3}{3} \right]$$

Simplifying,

$$= \left(\frac{1}{4} - 4 \right)$$

$$= -\frac{15}{4} \text{ units}$$

So, the correct answer is $-\frac{15}{4}$ units option B.

5. The area bounded by the curve $y = x|x|$, x-axis and the ordinate $x = 1$ and $x = -1$ is given by (Hint $y = x^2$ if $x > 0$ and $y = -x^2$ if $x < 0$)

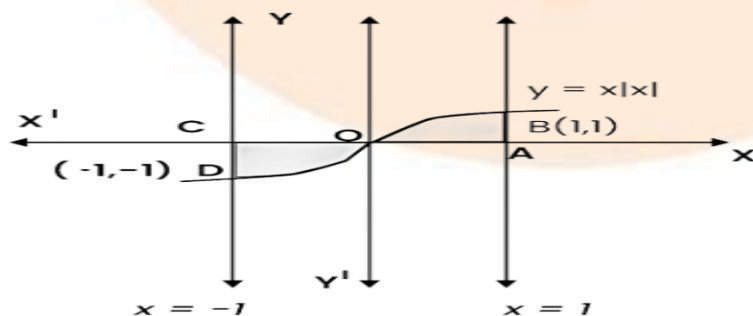
A. 0

B. $\frac{1}{3}$

C. $\frac{2}{3}$

D. $\frac{4}{3}$

Ans:



$$\text{Required area} = \int_{-1}^1 y dx$$

$$= \int_{-1}^1 x |x| dx$$

$$= -\int_{-1}^0 x^2 dx + \int_0^1 x^2 dx$$

Integrating using the power rule

$$= -\left[\frac{x^3}{3}\right]_{-1}^0 + \left[\frac{x^3}{3}\right]_0^1$$

Substituting the limits,

$$= -\left(0 - \frac{1}{3}\right) + \frac{1}{3}$$

$$= \frac{2}{3} \text{ units}$$

So, the correct answer is $\frac{2}{3}$ units option C.